Rudraksh Tiwari

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Summary

Final-year Computer Science student specializing in backend systems and correctness under concurrency. Built a distributed task queue with hand-written atomic job claim, crash recovery, and dead-letter handling, and a full-stack finance application deployed on AWS with per-user data isolation. Comfortable working across both Python/FastAPI and Java/Spring Boot ecosystems. Seeking SDE or Backend Engineer roles where I can own systems end to end.

Education

Nitte Meenakshi Institute of Technology, Bengaluru 2023 – 2027
Bachelor of Engineering – Computer Science & Engineering *CGPA: 8.32 / 10.0*

Deva Matha Central School 2023
Class 12 – Pre-University (CBSE) *Score: 76.6%*

Technical Skills

Languages: Java, Python, SQL, TypeScript
Backend: Spring Boot, FastAPI, PostgreSQL, Redis, REST APIs
Frontend: React, TypeScript, TailwindCSS, Streamlit
Infra & ML: Docker, AWS EC2, nginx, Git, scikit-learn, XGBoost, SHAP

Projects

Reliable Distributed Task Queue | *Python · Redis · Docker · FastAPI* | GitHub | *In Progress* 2025

- Building a Celery-style background job system from first principles – atomic job claim, crash recovery, retries with backoff, and a dead-letter queue all hand-written rather than delegated to a library.
- Implemented a Redis-list FIFO queue using atomic `LPUSH/BRPOP` for lossless, concurrent-safe enqueue/dequeue, with a `@task` registry and end-to-end producer/worker loop.
- Targets the guarantee that an accepted job completes even if a worker is killed mid-execution; staged roadmap through observability and chaos testing.

Clarifi: AI Financial Copilot | *FastAPI · React · PostgreSQL · ChromaDB · AWS* | Live | GitHub 2026

- Full-stack finance app that ingests real bank statement exports – dynamic transaction-header detection, separate Dr/Cr direction column, and day-first date parsing.
- Rebuilt a flawed Isolation Forest detector as log-scaled per-category robust z-scores (median/MAD), cutting 13 false positives to 5 explained flags on the same statement.
- Per-user RAG chat via sentence-transformers embeddings in ChromaDB with JWT auth and full data isolation; deployed on AWS EC2 behind nginx with TLS.

Loan Risk Analyzer | *XGBoost · SHAP · Streamlit* | GitHub 2024

- XGBoost credit-risk model trained on 45,000+ loan records, achieving ˜0.978 AUC-ROC with class-imbalance correction via `scale_pos_weight`.
- SHAP-based per-applicant risk drivers served through a Streamlit app with a three-tier Approved / Review / Rejected decision engine.

Experience

Web Development Intern Oct 2024 – Nov 2024
SkillCraft Technology *Bengaluru, Karnataka*

- Completed a 1-month internship building frontend and backend features with HTML, CSS, and JavaScript, working within real project structure, version control practices, and team development workflows.

Certifications & Achievements

Infosys Springboard – Machine Learning Foundation (2026); Citizen Data Science using Python, Software Engineering Fundamentals (2025); Java Foundation (2024)
MongoDB – Python Developer Path; Database Admin Path, Self-Managed (2025)
Coding Battles – Runner-Up (2024)